

Calm Months, Crash Months: Volatility-Managed Factors in Real Time and After Publication

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Abstract

Volatility-managed factors look strong in spanning regressions, but those regressions describe a portfolio no investor could have formed in advance. We rebuild the Fama-French five factors and momentum so that each month's exposure uses only prior data, is capped at 1.5 times and pays trading costs, and we estimate variance from every daily return with a one-month half-life, a learning rate chosen on pre-2017 data from 24 variants in a pre-registered comparison. From July 1973 to August 2026, only momentum improves significantly, from 0.43 to 0.94. After the published samples end, the canonical prior-month rule adds nothing to momentum, while the daily-data rule keeps part of the gain at half the turnover; over 116 months neither difference is statistically significant. Choosing the learning rate by recent performance does worse than fixing it, and no real-time combination beats its original factor after costs. An exact decomposition of the Sharpe ratio into a return channel and a risk channel explains the pattern.

Keywords: volatility-managed portfolios, factor timing, momentum, learning rate, out-of-sample performance

JEL: G11, G12, G14

1 Introduction

Moreira and Muir (2017) show that scaling a factor's exposure by the inverse of its recent variance raises the Sharpe ratio of many equity factors. Their evidence is a spanning regression of the managed factor on the original. A positive intercept means some mix of the two beats the original, but the mix uses full-sample moments no investor had. Cederburg et al. (2020) make the point directly: across 103 strategies ending in December 2016, real-time combinations of managed and original portfolios usually fail to beat the original. Liu et al. (2019) show that the market-timing version relies on a scale constant fitted with future data, and Barroso and Detzel (2021) that transaction costs erase most of the gains, with the market and momentum partly robust. Momentum is the most consistent survivor, in line with the crash evidence of Barroso and Santa-Clara (2015) and Daniel and Moskowitz (2016).

This letter asks three questions. Which of the six most widely used equity factors improve once every position uses only past data, leverage is capped and costs are paid? What happened after the

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published samples ended? Moreira and Muir’s data end in 2015 and Cederburg et al.’s in 2016, so January 2017 to August 2026 is out of sample for both, and McLean and Pontiff (2016) find that anomaly returns are 58% lower after publication. And does the rule stop adapting? Its estimators learn at fixed or vanishing rates; we test whether other learning rates keep it working.

The answers are narrow. Only momentum’s managed version beats the original over the full sample. After the published samples end, the canonical rule, which scales by the prior month’s realized variance, loses momentum’s gain, while a rule that uses every daily return with a one-month half-life keeps part of it at half the turnover; neither difference is statistically significant over a decade. Profitability, whose managed version earns a significant spanning alpha in Cederburg et al. (2020), shows no significant real-time gain. An exact decomposition into a return channel and a risk channel explains the pattern.

2 Data and method

We use the market, size (SMB), value (HML), profitability (RMW), investment (CMA) and momentum factors of Fama and French (2015) and Carhart (1997) from the Kenneth R. French Data Library, July 1963 to August 2026. Monthly returns f_m come from the monthly files; daily returns r_d serve only to measure risk. The canonical signal of Moreira and Muir (2017) is the prior month’s realized variance,

$$\text{RV}_m = \sum_{d \in m} (r_d - \bar{r}_m)^2, \quad (1)$$

where $\bar{r}_m$ is the mean daily return in month m . Our main signal uses every daily return with an exponential learning rate:

$$\hat{\sigma}_d^2 = \lambda \hat{\sigma}_{d-1}^2 + (1 - \lambda) r_d^2, \quad \lambda = 2^{-1/21}, \quad S_m = 21 \hat{\sigma}_{d(m)}^2, \quad (2)$$

where $d(m)$ is the last trading day of month m . A 21-day half-life puts about half the weight on the latest month and keeps some memory of earlier ones. It is one of the 24 variants of Section 4, chosen for momentum on 1973 to 2016 data under a plan committed before any variant was run; we apply it to all six factors so that the paper reports one rule, and report the canonical rule, $S_m = \text{RV}_m$, alongside. French’s daily momentum factor is re-formed daily, so for momentum both signals proxy the variance of the monthly-formed factor. The data are today’s vintage, which French revises as CRSP and Compustat update, not the returns as first published.

Managed factor. The exposure held in month m is set at the end of month $m - 1$ (Figure 1):

$$w_m = \min\left(\frac{c_{m-1}}{S_{m-1}}, L\right), \quad c_{m-1} = \frac{\text{sd}(f_1, \dots, f_{m-1})}{\text{sd}(f_s/S_{s-1} : s \leq m-1)}. \quad (3)$$

The scale c_{m-1} is the expanding-window version of Moreira and Muir’s full-sample constant: it makes the uncapped managed factor as volatile as the original over the data seen so far. The first 120 months are a burn-in, so positions start in July 1973 (638 months). The cap is $L = 1.5$, the higher of the two caps (1 and 1.5) in Moreira and Muir’s Table IV. Changing exposure costs $\kappa = 14$

basis points per unit, the highest of their cost levels, so the managed return is

$$f_m^\sigma = w_m f_m - \kappa |w_m - w_{m-1}|. \quad (4)$$

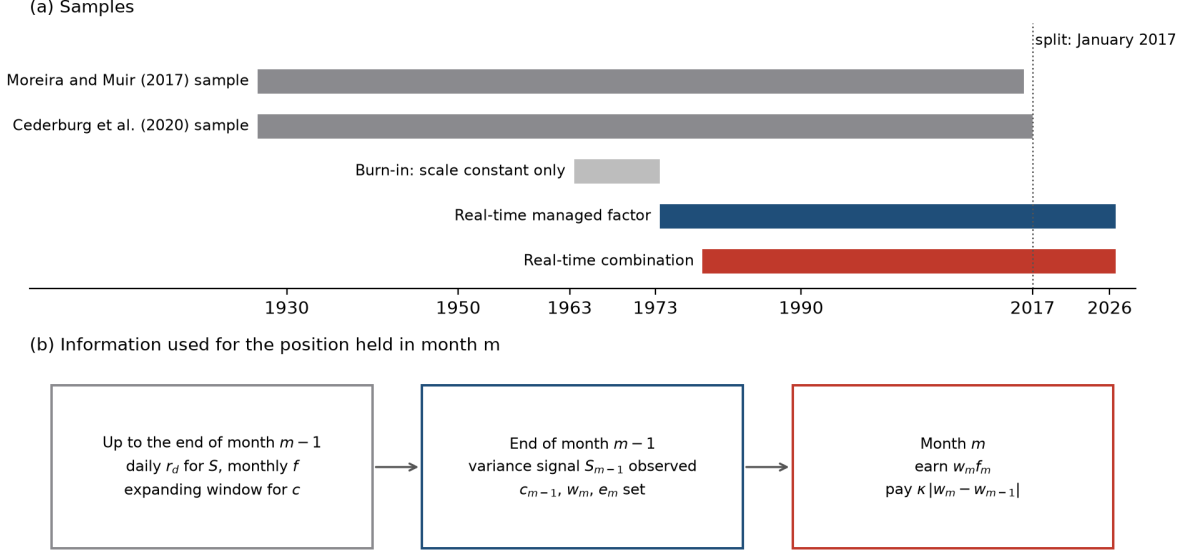


Figure 1: Study design. Panel (a): sample windows. The published samples start in the late 1920s for the market, size, value and momentum factors and in 1963 for profitability and investment; Moreira and Muir’s end in 2015 and Cederburg et al.’s in December 2016. Our first ten years only feed the scale constant. Panel (b): the position held in month m uses only information available at the end of month $m-1$.

Spanning regression. $f_m^\sigma = \alpha + \beta f_m + \varepsilon_m$, with Newey-West t -statistics (six lags). Because f^σ is net of costs, α is the return a static position in the original could not replicate, after costs.

Real-time combination. Following Cederburg et al. (2020), from month 60 of the real-time sample the investor forms the mean-variance mix of the original and managed factor using the expanding window $s < m$:

$$\begin{pmatrix} a_m \\ b_m \end{pmatrix} = \hat{\Sigma}_{m-1}^{-1} \hat{\mu}_{m-1}, \quad e_m = \text{clip}(k_m (a_m + b_m w_m), -L, L), \quad (5)$$

where $\hat{\mu}$ and $\hat{\Sigma}$ are the mean and covariance of (f_s, f_s^σ) and k_m scales the mix to the original factor’s volatility. Since $f^\sigma = w f$ before costs, the mix is a single exposure e_m to the original factor, capped like the managed factor and charged the same cost: $f_m^c = e_m f_m - \kappa |e_m - e_{m-1}|$.

Inference. For two return series with sample moments $\hat{v} = (\hat{\mu}_1, \hat{\mu}_2, \hat{\gamma}_1, \hat{\gamma}_2)$, where $\hat{\gamma}_i$ is the mean of squared returns, the Sharpe difference is

$$\hat{\Delta} = g(\hat{v}) = \frac{\hat{\mu}_1}{\sqrt{\hat{\gamma}_1 - \hat{\mu}_1^2}} - \frac{\hat{\mu}_2}{\sqrt{\hat{\gamma}_2 - \hat{\mu}_2^2}}, \quad \text{se}(\hat{\Delta}) = \sqrt{\nabla g(\hat{v})^\top \hat{\Psi} \nabla g(\hat{v}) / T}, \quad (6)$$

with $\hat{\Psi}$ a Bartlett long-run covariance of the four moment series, with lag length equal to the bootstrap block length. We test $\Delta = 0$ with the studentized circular block bootstrap of Ledoit and Wolf (2008): 4,999 resamples of 12-month blocks, each with its own block-based standard error se^* . The p -value is $(1 + \#\{|\hat{\Delta}^* - \hat{\Delta}|/\text{se}^* \geq |\hat{\Delta}|/\text{se}\})/5000$, and the 95% interval is $\hat{\Delta} \pm q_{0.95}^* \text{se}$, where $q_{0.95}^*$ is the 95th percentile of the resampled statistic. Within each family of six factors, p -values are adjusted with the step-down procedure of Holm (1979): with ordered $p_{(1)} \leq \dots \leq p_{(6)}$, the adjusted value is $\tilde{p}_{(j)} = \max_{i \leq j} \min\{1, (7 - i) p_{(i)}\}$. A test that the Sharpe gain differs across two periods treats the periods as independent and combines their bootstrap draws in the same studentized way.

Why timing can help or hurt. For any exposure series, the mean managed return decomposes exactly as

$$\overline{f^\sigma} = \bar{w} \bar{f} + \widehat{\text{cov}}(w_m, f_m) - \kappa |\overline{\Delta w}|, \quad (7)$$

with the covariance taken over the same months. The first term is what a constant exposure $\bar{w}$ would earn, and a constant exposure has the original factor's Sharpe ratio. Dividing the managed Sharpe ratio by the original's therefore gives, whenever $\bar{f} > 0$,

$$\frac{\text{SR}(f^\sigma)}{\text{SR}(f)} = \underbrace{\left(1 + \frac{\widehat{\text{cov}}(w_m, f_m) - \kappa |\overline{\Delta w}|}{\bar{w} \bar{f}}\right)}_{M: \text{ return channel}} \times \underbrace{\frac{\bar{w} \text{sd}(f)}{\text{sd}(f^\sigma)}}_{R: \text{ risk channel}}. \quad (8)$$

M measures the return that timing adds, net of costs, relative to a constant exposure of the same average size; R measures how much volatility timing removes relative to that exposure. Moreira and Muir (2017) motivate $w \propto 1/\sigma^2$ with the mean-variance weight $w_m^* = \mu_m/(\gamma\sigma_m^2)$. If the expected return μ_m does not vary with variance, the covariance is zero in expectation and the whole gain comes through R . If μ_m is lower in turbulent months, as in the momentum crashes of Daniel and Moskowitz (2016), the covariance adds a second gain through M . If μ_m is higher in turbulent months, M falls below one and the rule scales down exactly when the factor pays.

3 Results

3.1 The full real-time sample

In Table 1, managed momentum raises the Sharpe ratio from 0.43 to 0.94 after costs (95% interval for the gain 0.19 to 0.83; $p = 0.004$, Holm-adjusted 0.023). The canonical rule gives 0.90. The two rules are not significantly different ($p = 0.351$), but the daily-data rule trades half as much: a monthly exposure change of 0.13 against 0.25, and 45 to 51% of the canonical turnover across the six factors. At equal volatility, momentum's largest loss falls from 76 to 23 percentage points, and its gain would survive costs of up to 258 basis points per unit of exposure change, about 18 times the assumed level.

No other factor improves significantly. Profitability is the instructive case: its managed version earns a spanning alpha significant at 1% in Cederburg et al. (2020), but in real time its net spanning alpha is 0.9% a year ($t = 1.7$) and its Sharpe ratio moves from 0.38 to 0.45 ($p = 0.605$, unadjusted), a gap between in-sample alpha and real-time gains that Cederburg et al. (2020) trace to unstable

Factor	Managed factor, net				Real-time combination, net			Largest loss, equal volatility	
	Original	Canonical	Daily-data	p (Holm)	Original	Combination	p (Holm)	Original	Managed
Market	0.51	0.51	0.48	1.000	0.57	0.43	0.506	-68	-42
Size (SMB)	0.21	0.12	0.13	1.000	0.10	0.02	0.137	-77	-85
Value (HML)	0.32	0.23	0.27	1.000	0.24	0.18	0.343	-79	-67
Profitability (RMW)	0.38	0.47	0.45	1.000	0.47	0.35	0.545	-51	-46
Investment (CMA)	0.45	0.34	0.41	1.000	0.37	0.30	0.506	-31	-47
Momentum	0.43	0.90	0.94	0.023	0.42	0.80	0.343	-76	-23

Table 1: Annualized Sharpe ratios, July 1973 to August 2026 (638 months), exposure capped at 1.5, costs of 14 basis points per unit of exposure change. Canonical is the prior-month realized-variance rule; daily-data is the main specification, equation (2). The combination and the largest loss use the daily-data rule. The combination starts in July 1978; its first column is the original factor over the same months. p -values are Ledoit-Wolf studentized block-bootstrap tests of equal Sharpe ratios against the original factor, Holm-adjusted across the six factors; bold marks adjusted $p < 0.05$. Largest loss is the maximum peak-to-trough decline of summed monthly returns, in percentage points, with each series scaled to the original factor’s volatility.

spanning regressions.

No real-time combination beats its original factor after adjusting for six tests. For momentum it rises from 0.42 to 0.80 ($p = 0.081$ unadjusted, 0.343 adjusted). Removing the cap lifts it to 0.84 ($p = 0.048$, unadjusted) and removing costs as well to 0.88 ($p = 0.033$), consistent with Cederburg et al.’s finding that momentum’s combination adds value before frictions.

3.2 After the published samples

Factor	1973 to 2016 (522 months)					2017 to 2026 (116 months)				
	Original	Canonical	Daily-data	p (Holm)	α , % (t)	Original	Canonical	Daily-data	p (Holm)	α , % (t)
Market	0.44	0.49	0.45	1.000	0.9 (0.93)	0.81	0.59	0.63	1.000	-0.1 (-0.03)
Size (SMB)	0.31	0.25	0.24	1.000	-0.1 (-0.15)	-0.25	-0.53	-0.45	1.000	-1.1 (-1.29)
Value (HML)	0.45	0.33	0.36	1.000	0.1 (0.18)	-0.12	-0.61	-0.46	1.000	-1.2 (-2.05)
Profitability (RMW)	0.42	0.51	0.50	1.000	1.0 (1.74)	0.26	0.27	0.16	1.000	-0.1 (-0.23)
Investment (CMA)	0.62	0.51	0.56	1.000	0.3 (0.69)	-0.11	-0.52	-0.46	1.000	-1.5 (-2.24)
Momentum	0.48	1.00	1.02	0.110	5.6 (4.89)	0.21	0.23	0.44	1.000	1.2 (1.78)

Table 2: Real-time managed factors (cap 1.5, net of costs) before and after the published samples end. Sharpe ratios are annualized; p -values test the daily-data rule against the original factor and are Holm-adjusted within each period; α is the daily-data rule’s annualized net spanning alpha with its Newey-West t -statistic.

Table 2 and Figure 2 split the sample at January 2017. Before the split, managed momentum’s gain is large (0.48 to 1.02; $p = 0.018$ unadjusted, 0.110 after adjusting for six factors in this shorter window). After it, the canonical rule gains nothing (0.23 against the original’s 0.21), while the daily-data rule keeps part of the gain (0.44; $p = 0.288$ unadjusted, 95% interval -0.29 to 0.76; against the canonical rule, $p = 0.101$). A test that the gain changed between the two periods gives $p = 0.242$: the decade is consistent both with no gain and with the earlier one.

Value’s post-2016 net alpha of -1.2% ($t = -2.05$) is not a finding: the Sharpe test does not reject ($p = 0.179$) and with a January 2016 split the alpha is -0.6% ($t = -0.99$).

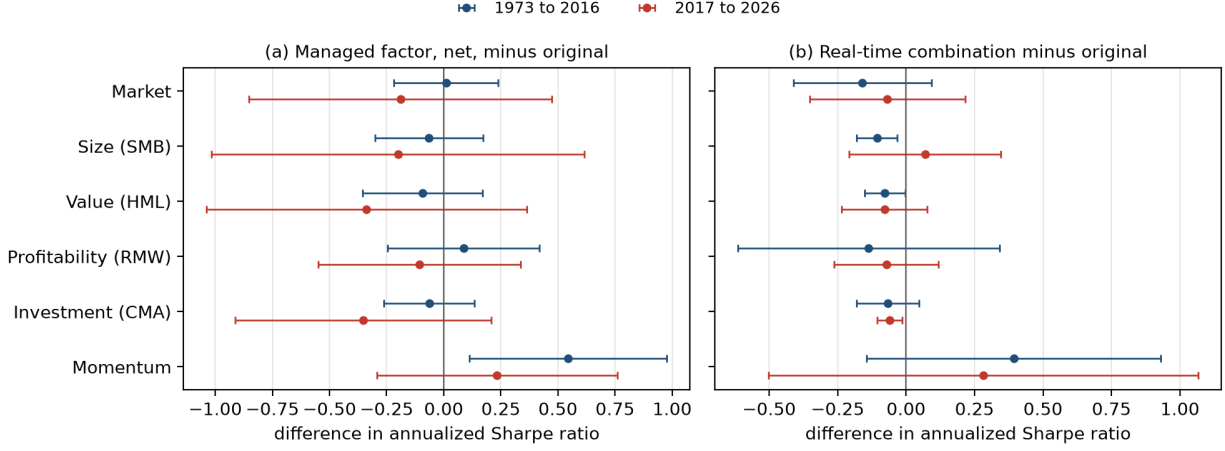


Figure 2: Sharpe ratio gain of the daily-data rule over the original factor, with 95% Ledoit-Wolf studentized bootstrap intervals, before and after the published samples end. Panel (a): managed factor, net of costs. Panel (b): real-time combination, net of costs.

3.3 Why: a return channel and a risk channel

Factor	Period	$\bar{w}$	$\bar{f}$	$\bar{w}\bar{f}$	$\widehat{\text{cov}}(w, f)$	Cost	$\bar{f}^\sigma$	M	R
Market	1973 to 2016	0.74	6.9	5.1	-0.2	0.2	4.6	0.91	1.13
	2017 to 2026	0.73	13.1	9.6	-2.8	0.3	6.5	0.67	1.14
Size (SMB)	1973 to 2016	0.59	3.2	1.9	-0.3	0.2	1.4	0.71	1.10
	2017 to 2026	0.43	-2.6	-1.1	-0.9	0.1	-2.2	n/a	0.90
Value (HML)	1973 to 2016	0.79	4.5	3.6	-0.9	0.2	2.4	0.67	1.18
	2017 to 2026	0.27	-1.7	-0.4	-1.0	0.1	-1.5	n/a	1.08
Profitability (RMW)	1973 to 2016	0.84	3.4	2.8	-0.4	0.2	2.3	0.79	1.52
	2017 to 2026	0.40	2.4	1.0	-0.4	0.1	0.4	0.45	1.33
Investment (CMA)	1973 to 2016	0.95	4.2	4.0	-0.8	0.2	2.9	0.73	1.22
	2017 to 2026	0.50	-1.0	-0.5	-1.1	0.2	-1.8	n/a	1.11
Momentum	1973 to 2016	0.82	7.3	6.0	2.5	0.2	8.2	1.37	1.56
	2017 to 2026	0.35	2.9	1.0	1.0	0.1	1.9	1.87	1.13

Table 3: Exact decompositions of the daily-data rule, equations (7) and (8). $\bar{w}$ is the average exposure. $\bar{f}$, $\bar{w}\bar{f}$, the covariance, the cost and $\bar{f}^\sigma$ are annualized, in percent, and $\bar{w}\bar{f} + \widehat{\text{cov}} - \text{cost} = \bar{f}^\sigma$ up to rounding. M and R are the return and risk channels; their product is the ratio of the managed to the original Sharpe ratio. M is not reported when $\bar{f} \leq 0$.

Table 3 applies equations (7) and (8). Before 2017, both channels lift momentum: the covariance adds 2.5 percentage points a year against 0.2 of costs ($M = 1.37$), and cutting exposure in volatile months removes about a third of the risk of a constant exposure ($R = 1.56$). After 2016, momentum's own return fell from 7.3% to 2.9% a year and its variance stayed high relative to its history, so exposure was low (average 0.35, below 0.2 in 36 of 116 months). The daily-data rule still earned a covariance of 1.0 points at 0.1 of costs ($M = 1.87$, $R = 1.13$). The canonical rule earned 0.4 points and spent 0.3 on trading. Value's covariance is negative in both periods (-0.9 and -1.0 points), so the rule works against it. For profitability before 2017, the return channel ($M = 0.79$) gives back

part of what the risk channel adds ($R = 1.52$), leaving a gain too small to be statistically significant.

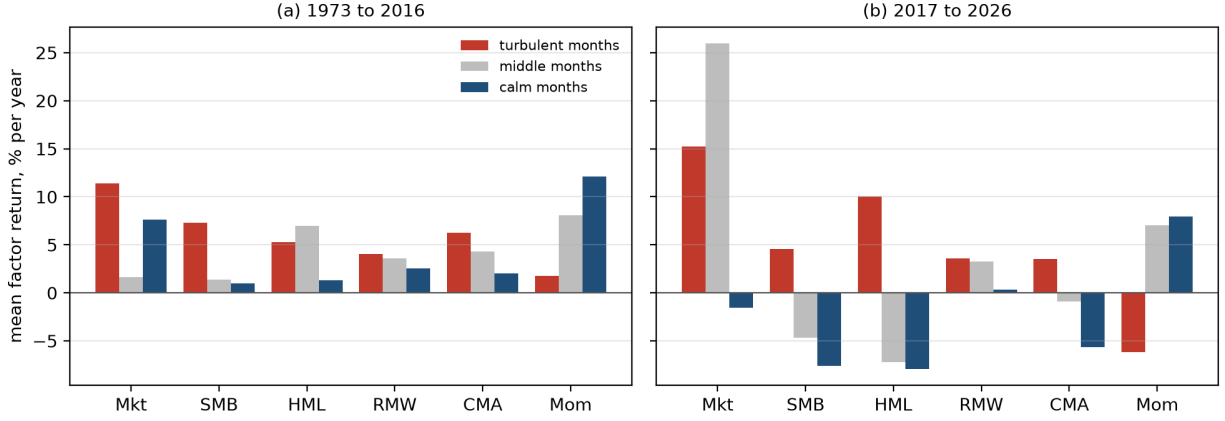


Figure 3: Annualized mean factor return by tercile of real-time managed exposure (daily-data rule). Turbulent months are the third with the lowest exposure, that is, the highest prior variance.

Figure 3 shows the covariance directly. Momentum earns most in calm months: after 2016, 8.0% a year in the calm tercile against -6.1% in the turbulent one. Value is the mirror image, earning 10.1% in turbulent months and -7.9% in calm ones. Figure 4 shows the consequence for value: its eight best months since 2017, 7 of them in the 2021 to 2022 rebound, came while the managed strategy held between 0.06 and 0.37 times the factor.

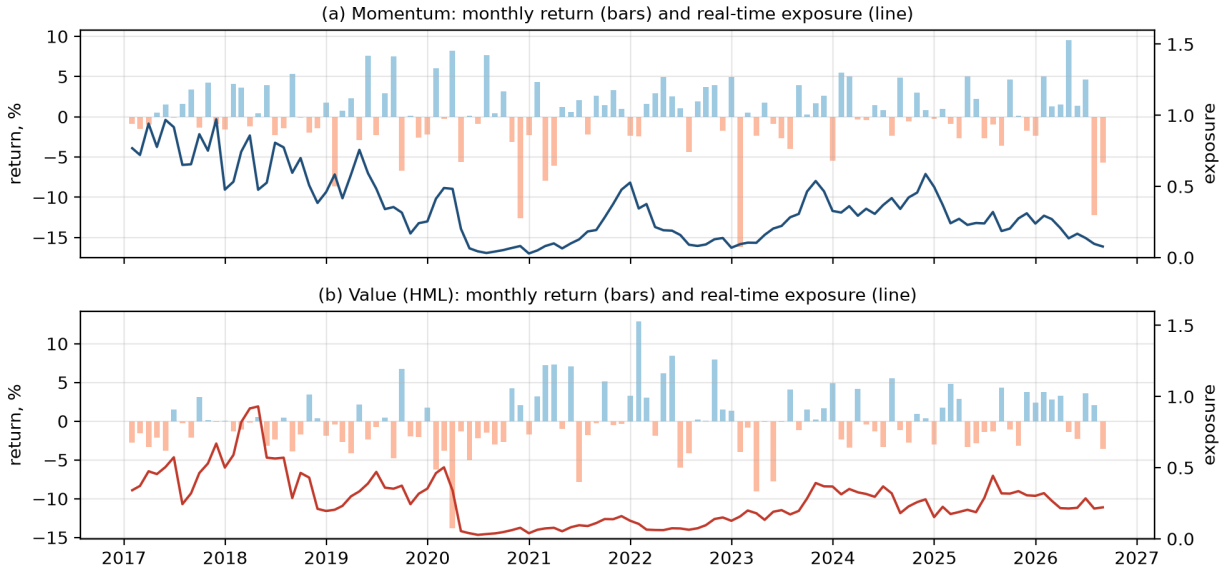


Figure 4: Monthly factor return (bars) and real-time managed exposure (line, daily-data rule, cap 1.5), January 2017 to August 2026.

Specification	Momentum, full sample			Value, after the split		
	Original	Managed	p	α , %	t	p
Baseline	0.43	0.94	0.004	-1.2	-2.05	0.179
Canonical signal (prior-month RV)	0.43	0.90	0.005	-1.6	-2.74	0.065
6-month bootstrap blocks	0.43	0.94	0.002	-1.2	-2.05	0.127
24-month bootstrap blocks	0.43	0.94	0.004	-1.2	-2.05	0.287
Split at January 2016	0.43	0.94	0.004	-0.6	-0.99	0.412
Cap 2	0.43	0.97	0.003	-1.2	-2.05	0.179
No cap	0.43	0.97	0.003	-1.2	-2.05	0.179
Inverse volatility	0.43	0.78	0.014	-1.3	-1.76	0.185
Start in July 1978	0.42	0.89	0.013	-1.2	-2.05	0.179

Table 4: Robustness. Momentum columns test the managed against the original factor over the full real-time sample (the last row starts in July 1978); value columns report the net spanning alpha and the Sharpe test after the split (January 2017 unless stated). p -values are unadjusted. Value’s exposure never exceeds 1.5 after 2016, so its row does not change across caps.

3.4 Robustness

Table 4 varies the variance signal, the bootstrap block length, the split date, the cap, the scaling rule and the start date. Managed momentum’s full-sample Sharpe ratio ranges from 0.78 to 0.97 and every test rejects equality with the original ($p \leq 0.014$). The burn-in needs no check: the scale constant uses every month since 1963 whatever its length, so it only sets when trading starts. Figure 5 shows momentum’s path at equal volatility.

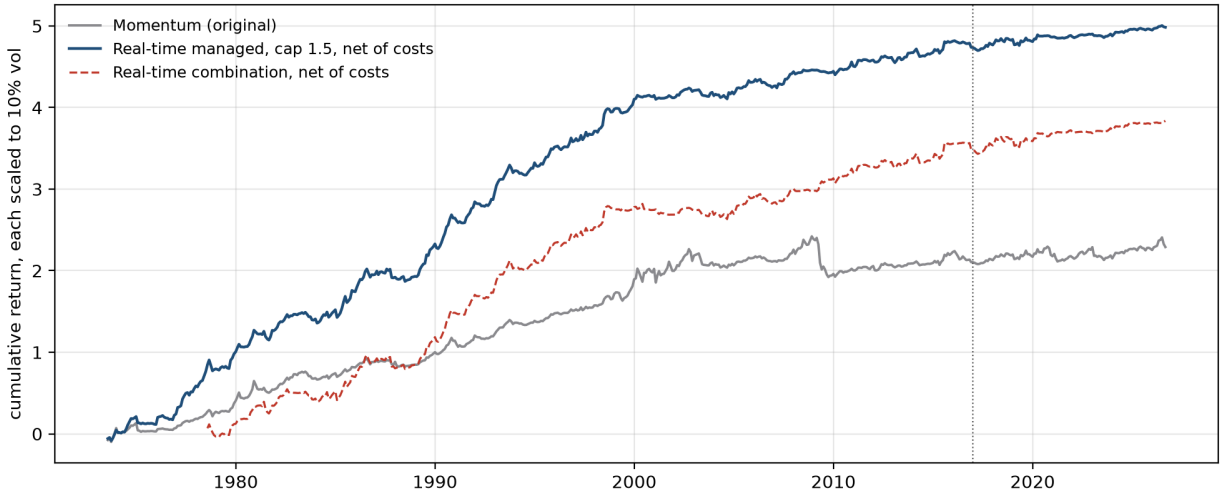


Figure 5: Cumulative summed monthly returns of momentum, its real-time managed version and the real-time combination (daily-data rule, both net of costs), each scaled to 10% annualized volatility for display. Scaling leaves every Sharpe ratio unchanged. The dotted line marks January 2017.

4 Learning rates

The rule has three estimators, each with an implicit learning rate. The canonical signal weights the last month’s days equally and ignores earlier ones; the scale constant c and the combination weights use expanding windows, whose step size $1/t$ has fallen to about $1/640$ by 2017, so they have effectively stopped learning. After the canonical rule’s post-2016 results were known, and before any alternative was run, we committed a plan (`paper/PLAN_adaptive.md` in the code repository) that fixes 24 variants: six signals (S0, prior-month realized variance; S1 to S3, exponentially weighted daily variance with half-lives of 5, 21 and 63 trading days; S4, a forecast in the spirit of Corsi (2009) from the last day, week and month; S5, realized variance of the last 13 weekly returns), four scale constants (B0, expanding; B1 to B3, half-lives of 3, 5 and 10 years) and two exponentially weighted combinations. Because the post-2016 results were already known, no rule uses them to pick a variant: the best variant on 1973 to 2016, evaluated once afterwards; a real-time rule holding each month the variant with the best trailing 10-year Sharpe ratio; and the superior-predictive-ability test of Hansen (2005) against the canonical rule, with loss differentials $r_{k,m}/\text{sd}(r_k) - r_{0,m}/\text{sd}(r_0)$, whose mean is the monthly Sharpe difference. The pre-committed adoption rule made the pre-2017 selection for momentum the main specification if either of the last two showed a significant improvement for momentum with the same sign after 2016.

Factor	Selected on 1973 to 2016	2017 to 2026				Real-time selection, from July 1983			Hansen test p	
		Original	Canonical	Selected	p	Canonical	Selected	p	Full	2017 to 2026
Market	S0-B1	0.81	0.59	0.65	0.131	0.60	0.63	0.557	0.466	0.151
Size (SMB)	S5-B1	−0.25	−0.53	−0.34	0.188	−0.21	0.00	0.018	0.409	0.301
Value (HML)	S4-B1	−0.12	−0.61	−0.21	0.008	0.18	0.14	0.573	0.147	0.008
Profitability (RMW)	S3-B0	0.26	0.27	0.14	0.241	0.66	0.64	0.836	0.561	0.822
Investment (CMA)	S3-B1	−0.11	−0.52	−0.36	0.323	0.21	0.28	0.252	0.082	0.088
Momentum	S2-B0	0.21	0.23	0.44	0.101	0.82	0.70	0.019	0.504	0.045

Table 5: Learning-rate selection. Variant codes: signal S0 to S5 and scale constant B0 to B3 (Section 4); S0-B0 is the canonical rule and S2-B0 the daily-data rule. p -values in columns six and nine are Ledoit-Wolf tests against the canonical rule, unadjusted. Hansen’s test asks whether any of the 23 other variants beats the canonical rule.

For momentum the pre-2017 selection is S2-B0 (Sharpe ratio 1.02 against 1.00 for the canonical rule before 2017), which afterwards earns 0.44 against 0.23 ($p = 0.101$). Hansen’s test finds no variant better than the canonical rule over the full sample ($p = 0.504$) but rejects after 2016 ($p = 0.045$, and between 0.028 and 0.069 for block lengths of 6 to 24 months). That met the adoption rule, at the margin. The real-time rule points the other way: chasing the recently best learning rate switched variant 95 times and earned 0.70 against 0.82 for the canonical rule over the same months ($p = 0.019$). The learning rate that worked best recently is not the one that works next.

Figure 6 shows three further patterns. The fastest signal, a 5-day half-life, trails the canonical rule before 2017 for 5 of the six factors, but helps in a crash. At the end of February 2020 it had cut market exposure to 0.49, against 0.69 for the canonical rule, and its net return over February to April was -5.3% instead of -8.1% (Figure 7), although no signal held more than 0.07 of the market for the April rebound. Weekly realized variance over a quarter is the best momentum variant after 2016 (0.53) but among the worst before (0.94). With the canonical signal, a 3-year half-life for the

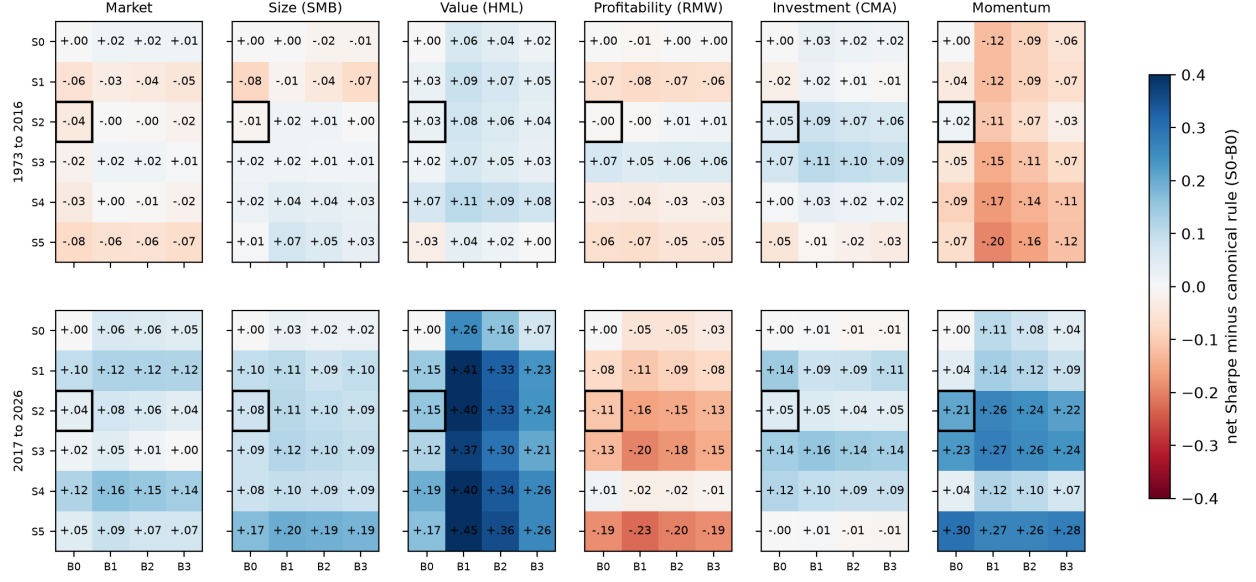


Figure 6: Net Sharpe ratio of each of the 24 pre-registered variants minus the canonical rule (S0-B0), by factor, before (top) and after (bottom) the published samples end. The outlined cell is the daily-data rule (S2-B0).

scale constant lowers momentum's pre-2017 Sharpe ratio to 0.88 but raises its post-2016 one to 0.34. An adaptive scale constant helps value most (column B1 in Figure 6): its pre-2017 choice, S4-B1, cut value's post-2016 loss against the canonical rule (Sharpe ratio -0.21 against -0.61 , $p = 0.008$; Hansen $p = 0.008$), the clearest case for a faster learning rate. Exponentially weighted combination weights do worse than expanding ones for 6 of the six factors. Faster learning trades stability for recent fit, and the trade rarely pays. Faster rebalancing is not tested: every French daily momentum series is re-formed daily, so daily returns of the monthly-formed factor would need CRSP.

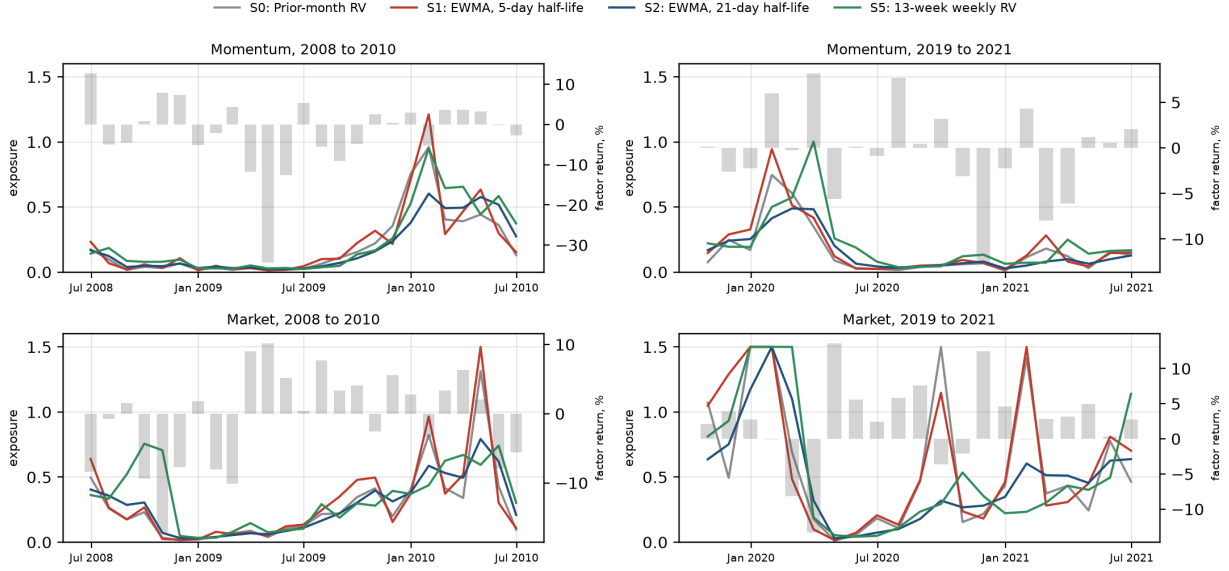


Figure 7: Real-time exposure under four variance signals (lines, expanding scale constant) and the factor's monthly return (bars) around the 2008 to 2009 and 2020 crises.

5 Conclusion

Built with only past data, capped leverage and costs, volatility management improves one of six widely used equity factors, momentum. After the published samples end, the canonical rule stopped adding value to momentum; estimating variance from every daily return with a one-month half-life keeps part of the gain at half the turnover. A decade is too short to show that this, or any learning rate, is reliably better, and choosing the learning rate by recent performance is worse than fixing it. No real-time combination beat its original factor after costs. Equation (8) gives investors a check before adopting the rule: it pays only when the product of the return and risk channels exceeds one.

Two limits apply. The post-2016 window has 116 months, enough to detect a large change but not a moderate one. We charge costs only on changes in exposure, not on the trading inside each factor portfolio, so every net figure is an upper bound.

Data availability. Factor returns are publicly available from the Kenneth R. French Data Library. Code is available from the author on request.

Declaration of generative AI and AI-assisted technologies in the writing process. During the preparation of this work the author used generative AI tools to create the simulations and code and to draft the content structure of the paper. The author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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